

Game Demo

Exercise 1: Use the bullet physics engine within Panda3D to develop a basic demo of a game idea.

Your demo does not have to be a complete game but should demonstrate some aspect of a game. Ideally it will be a novel idea, but you can also demonstrate a game mechanic that you have seen elsewhere. You are expected to use the bullet physics engine as a base but you can modify it if needed. We suggest focusing on some game mechanics for the player (similar to a tutorial level). I recommend looking at the questions below before deciding on what you do.

There should be some kind of achievement possible for the player: equivalent to scoring a goal, shooting a target, reaching a platform, operating a device, opening a door, etc... The demo should print a message or otherwise minimally inform the player of success.

This is just a very simple demo. You do not need to create any 3d models: you can use existing models from Panda3d, bullet samples or Alice.org (<https://www.alice.org/pandagallery/>).

You do not need to use any animated models in a 3d graphics sense (e.g. actual walking leg movement). The movement and behaviour only need to use the physics engine and player inputs. Do not attempt any features beyond the scope of the course such as networking, use of controllers, AI opponents or game components outside of Panda3d or bullet. If you wish to include a moving target, it is fine to have some simple movement such as a loop.

You can build from or combine any existing sample code from your panda3d install (e.g. samples), panda3d.org (e.g. Bullet Samples or manual code at <https://docs.panda3d.org/1.10>) and course material. You may not copy/paste code from outside Panda3d or bullet demos directly. You can look at it to see how to do something. You *must* indicate where any code and algorithms came from (in your hand-in).

Hand in your game demo as a complete zip to avenue that includes the python file (e.g. *gamedemo.py*) and any models you use in a single zipped folder. It should unzip and let someone play right away. You can put the PDF with answers to questions as a separate file on avenue (*project.pdf*).

Exercise 2: Provide (very brief) written instructions for how to play your demo and what the goal is. This should be sufficient to play it and achieve the goal.

Exercise 3: Describe the key physics (as taken from the bullet engine) that makes your game work. In particular, describe any choices and changes such as to gravity, friction, bouncing and so on.

Exercise 4: Describe any novel mechanics in your game. Novel mechanics are often what sets a game apart and attracts attention. This would include things that apply special forces (like a jump pad), flying or wind type mechanics that push the player, ways the player can manipulate / change the world and so on.

Exercise 5: Describe how your demo might be fleshed out into a full game. What components are still required? What category of game would it be? e.g. Do you see it as multiplayer, a phone game, a puzzle game, a competitive/e-sport type game, etc....

Exercise 6: Briefly critique your game demo (and the bullet engine effectively if that is holding you back). What physics do you like to have that bullet doesn't provide?

Exercise 7: Similar to the makers of Rocket League, do you see a need to change how physics works to make your final game more fun or more playable?